

PROFINET测试认证

闫菲 西门子数字化工业集团



SIEMENS

PROFINET 并不是简单的IO通讯



为什么PROFINET产品一定要通过测试认证

- ❑ 通过保证设备的一致性可以消除系统集成中令人头痛的兼容性问题
- ❑ 作为工业产品的供应商，提供通过测试认证的产品，给客户以合规、合法积极正面的形象。通过测试认证提高产品在客户中的信任度
- ❑ 没有通过测试的产品会把问题留给现场、留给客户。工程师出差救火的费用很高，而且在客户面前的形象也会变差
- ❑ **PI国际官方的说法：**
 - 对PROFINET设备进行测试认证是产品质量保证的重要组成部分
 - 测试认证保证了在现场运行的设备的一致性和互操作性
 - PROFINET的完整规范是设备应用必要组成部分，并且决定了从设备可访问到的信息
 - PROFINET设备的测试认证是必须的
 - 如果设备是驱动器和编码器，那么行规的测试（例如PROFIDRIVE）也是必须的



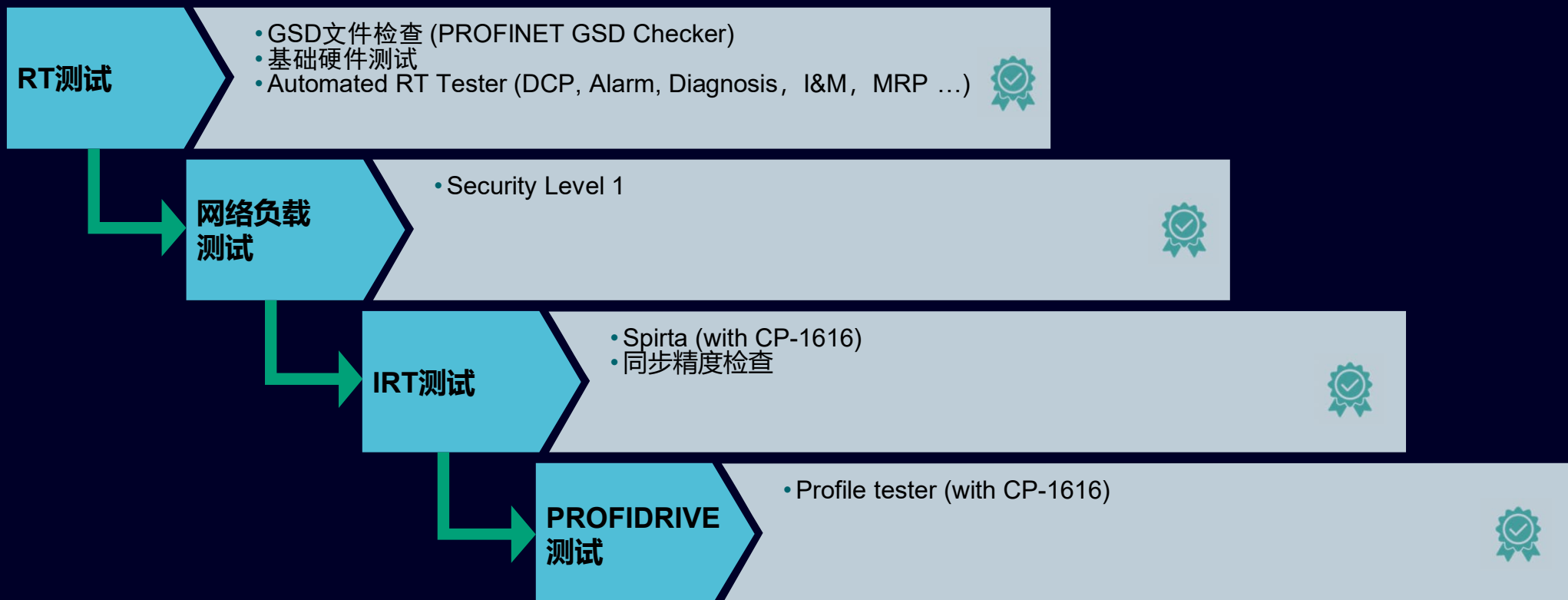
如何顺利通过测试

选型	- 从认证的供应商获取协议栈、芯片或模组 - 在PI官方网站查询开发包的认证信息（例如：EK-ERTEC200P-2） - 如果厂商的开发包没有测试证书，那么基于该开发包实现的产品大概率是没法通过测试的
研发	- 申请Vendor ID - 申请MAC地址 - 定制GSDML文件 - 适配I&M0-5, SNMP数据
预测试	- 西门子北京CoC申请预测试 - 或自行搭建测试环境（建议Profidrive行规部分自行测试） - 根据测试报告和指导进行修改 <i>耐心！</i>
PI测试认证	- 在PI授权实验室进行产品一致性与互操作性测试 - 向PI申请产品认证 - PI发布产品认证



IRT伺服驱动测试流程

PI会员可免费下载PROFINET测试工具软件及测试规范(Test Bundle)



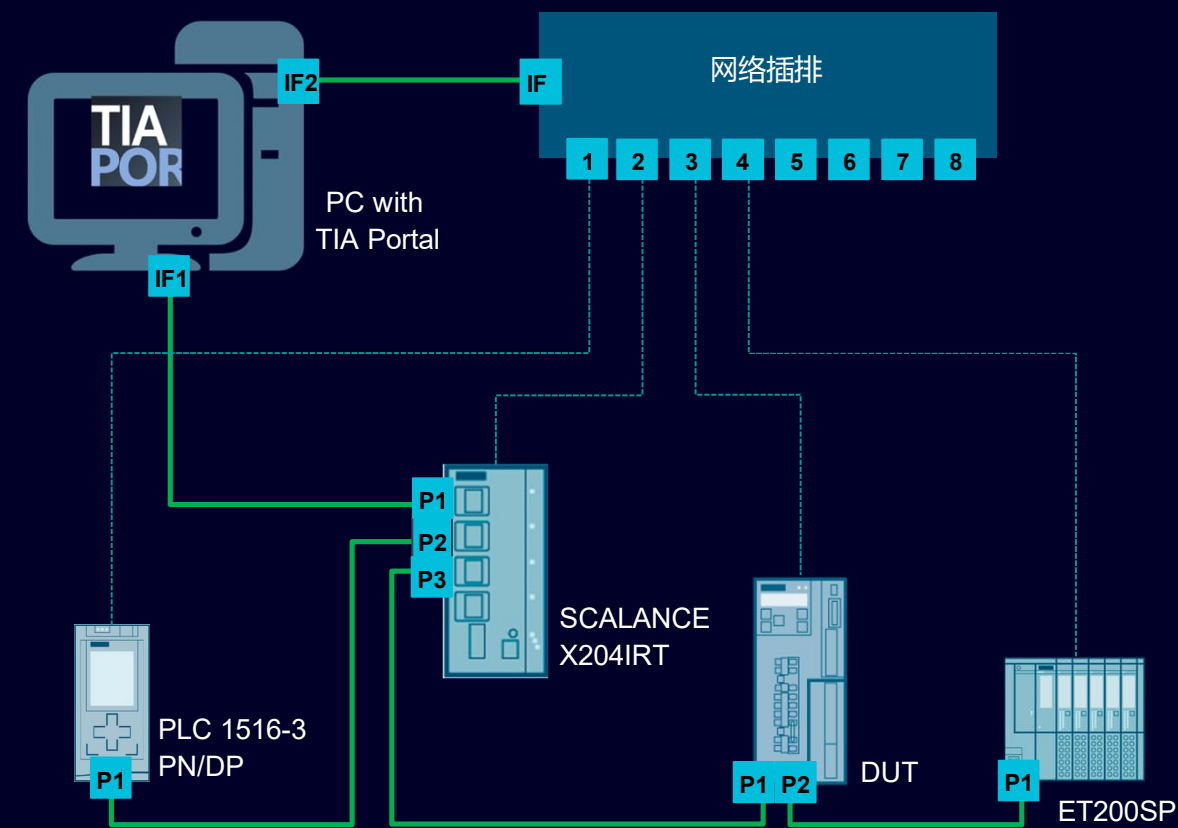
测试环境硬件需求

1	PLC 1516-3 PN/DP (device A)	?
2	ET200SP + DI&DQ (device D)	✓
3	网络插排	✓
4	SCALANCE X204IRT (device B)	✓
5	带PCI插槽PC机 , windows/ubuntu 双系统	✓
6	CP-1616 (for IRT and Profidrive)	?
7	10米网线 X 3	✓
8	I350-T4/T2 网卡	✓

被测设备信息要求

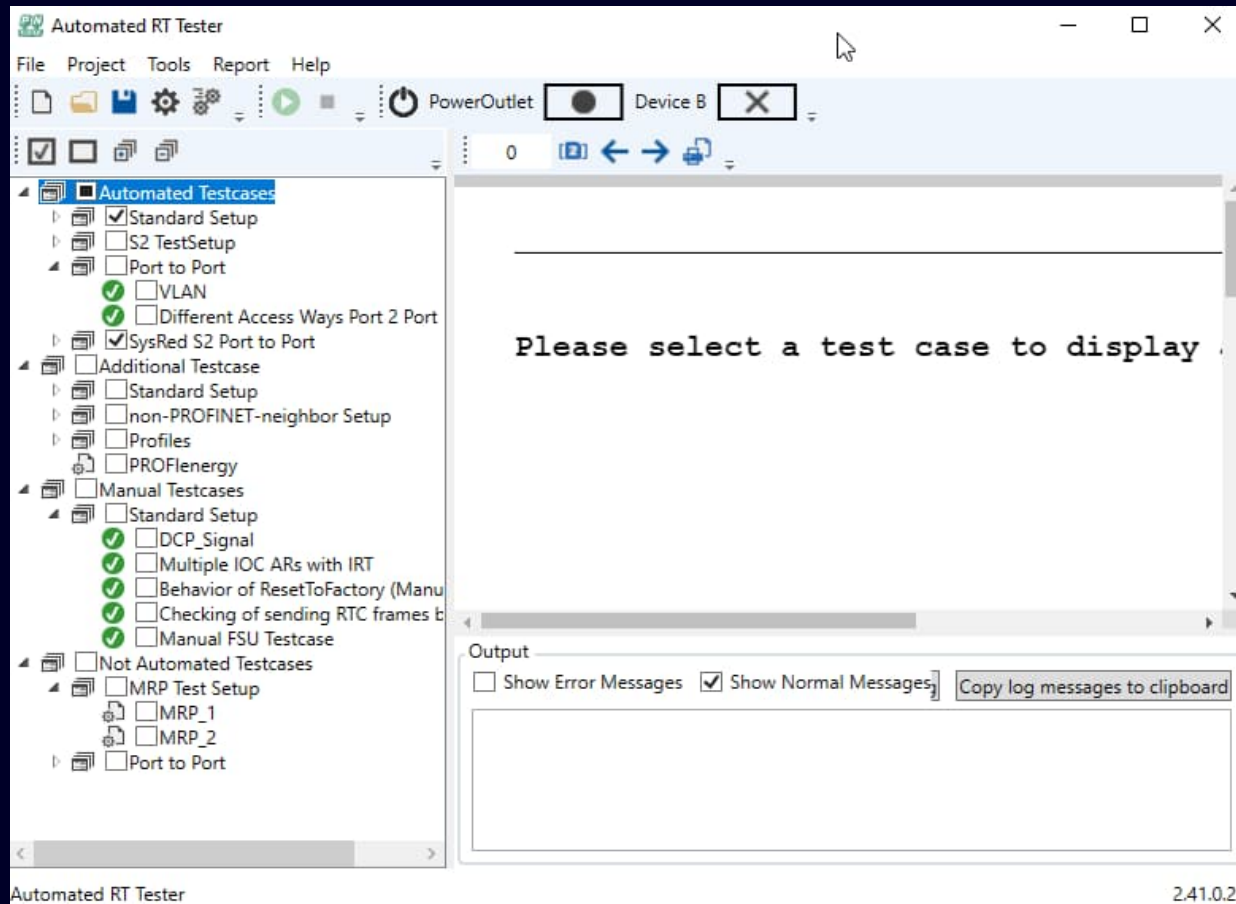
- ❑ 提供信息完整准确的设备 GSD 文件 (V2.4)
 - PROFINET_GSD_Checker_V2.42 软件
 - pn_upgrade_gsdml_V2.42 软件
 - GSDML-Spec_2352_V242_May21规范及示例文件
 - Vendor ID 及 MAC 地址等
 - RT / IRT 通讯类别
 - maxInput / maxOutput 1440
- ❑ 提供被测设备的产品说明书, 包括网口编号、连线方法及升级方法等
- ❑ 提供被测设备所需的特殊电源、电缆、接插件等
- ❑ 提供改变被测设备输入、观察被测设备输出变化的附加设备

Automated RT Tester 测试场景示例



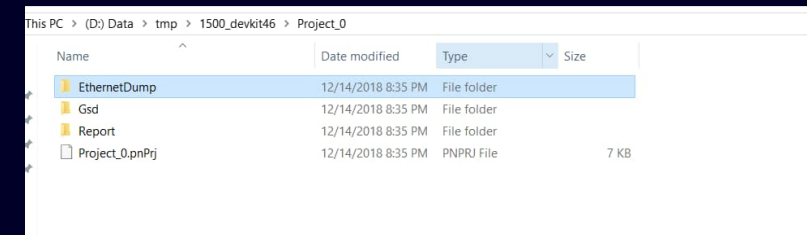
A	PLC 1516-3 PN/DP	192.168.0.100
B	SCALANCE X204IRT	192.168.0.99
DUT	PROFINET被测设备	192.168.0.50
D	ET200SP + DI&DQ	192.168.0.98
	PC机, I350-T4/T2 网卡	192.168.0.25
	网络插排	
	10米网线 X 3	

测试软件示例 – Automated RT Tester

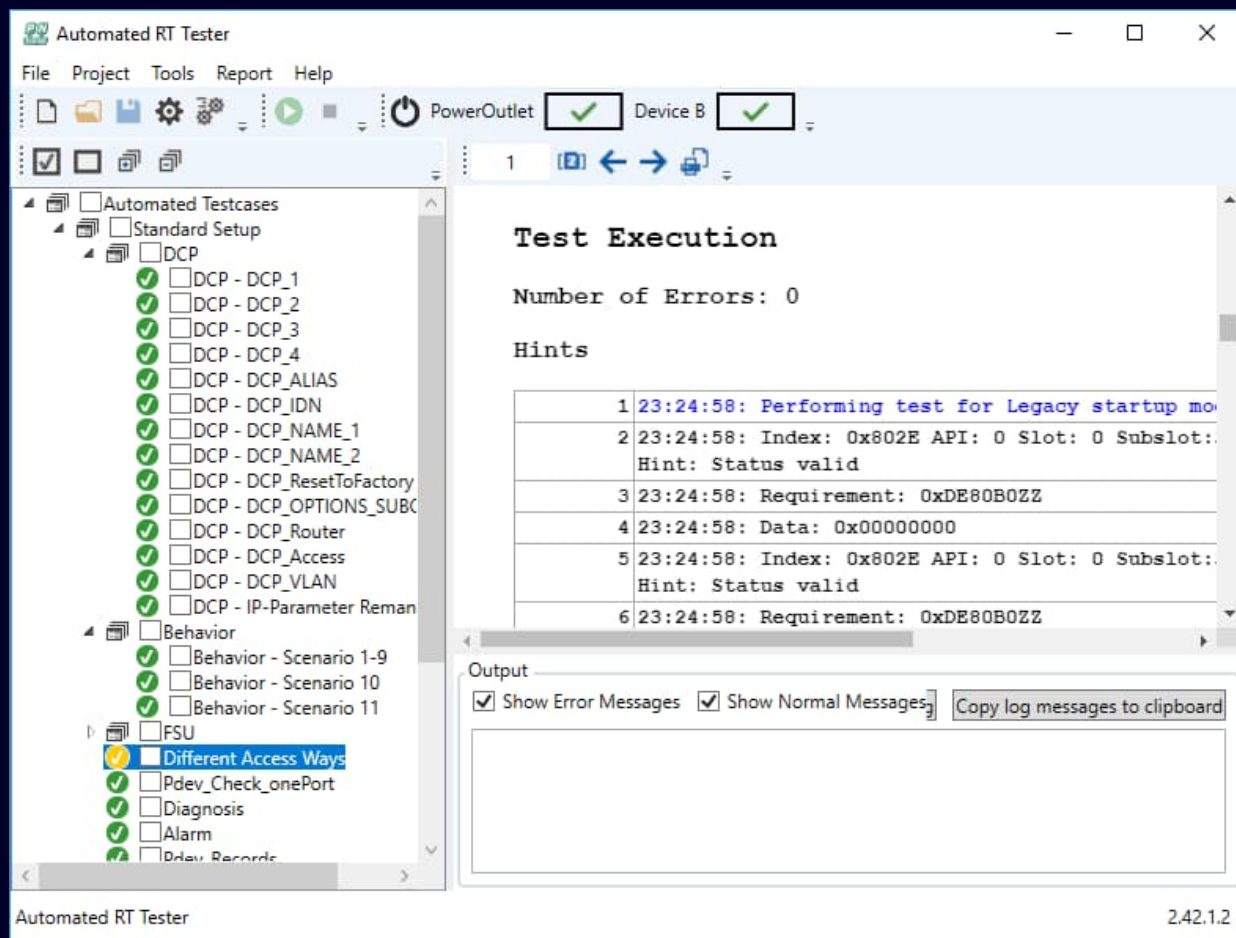


Automated RT Tester:

- 测试Profinet协议栈一致性
- 测试有自动和手动两部分
- 测试过程中可能会改变拓扑
- 测试过程抓包可供分析



测试软件示例 – Automated RT Tester



- 根据测试提示错误的项目，找到对应的 test case 文档并查阅错误原因。
- 根据测试过程抓包文件进行分析

TestCaseSpec > device_RT > automated

名称	修改日期
TCS_00000001_DiffAccessWays_V2.41.0.2.pdf	2020/11/10 8:50

8.3.26 Read Index 0x802E

Pass:

- This is a "reserved" Index (legacy). It shall be responded with "Access – invalid index".
- If it responds with OK, Pass with hint!

Inconclusive:

- ---

Fail:

- Check is not according to the expected values mentioned in "pass".

File Name	Date	Size
DCP_-_DCP_VLAN.pcap	2022/9/27 19:16	66 KB
DCP_-_IP-Parameter_Remanence.pcap	2022/9/27 19:23	92 KB
Diagnosis.pcap	2022/9/28 11:54	1,242 KB
Different_Access_Ways.pcap	2022/9/28 11:26	627 KB
Different_Access_Ways_Port_2_Port....	2022/9/28 12:29	1,961 KB
IP_UDP_RPC_I&M_EPM.pcap	2022/9/28 12:22	199 KB

网络负载测试 – SecurityLevel1Test

```
root@pnio: ~/Netload/Scripts
File Edit View Search Terminal Help
Some of the Netload generating sequences have to be checked.
To allow this the script starts the application dumpcap
and begins a recording.
The recordings of this testrun will be stored in the folder root/Netload/Out
and have to be checked by the user after the Netload sequence has finished.
Press enter to continue.
=====
Select the number of ports on the DuT!
=====
(1) PN Device: 1 port RT (Cu and FO)
(2) PN Device: 2 or more ports RT (Cu and FO) or IRT (Cu)
(3) PN Device: 2 or more ports Device IRT (FO)
(4) PN Controller: 1 port RT (Cu and FO)
(5) PN Controller: 2 or more ports RT (Cu and FO) or IRT (Cu and FO)
=====
2
Number of ports (2) PN Device: 2 or more ports RT (Cu and FO) or IRT (Cu) select
ed!
=====
Select the Data Rate for the load generating Interface!
=====
(0) 100 Mbit
(1) 1 Gbit
(exit) Exit
=====
0
Interface (0) 100 Mbit selected!
=====
Select the Netload Class to be applied to the sequences!
=====
(0) Netload Class I
(1) Netload Class II
(2) Netload Class III
(3) Return to menu 'Data Rate'
(exit) Exit
=====
2
(2) Netload Class III selected!
=====
Select the Communication Quality to be tested!
=====
+++ N E T L O A D - Generation
+++ (Load profile, each 1 minute per frame) max. 2 hours
+++++
(0) Faulty Communication
(1) Normal Communication
(2) Return to menu 'Netload Class'
(exit) Exit
```

SecurityLevel1Test :

- Faulty Communication
- Normal Communication
- 示例PLC程序

8.3.2.2 PROFINET Device Test

Normal operation:

Pass:

- First run shall pass or a minimum three out of five runs are done without errors.
- AR shall stay established.
- Status of data shall be good.
- Actual (RT or IRT service), and acyclic services shall work in a proper manner.
- The behavior of the DUT is according to the test cases (8.3.3 -8.3.83)

Fail:

- If an AR breakdown occurs with testing **normal** operation
- The DUT resets itself
- A reset is necessary to start the DUT again
- The behavior of the DUT is not according to the test cases (8.3.3 -8.3.83)

Faulty operation:

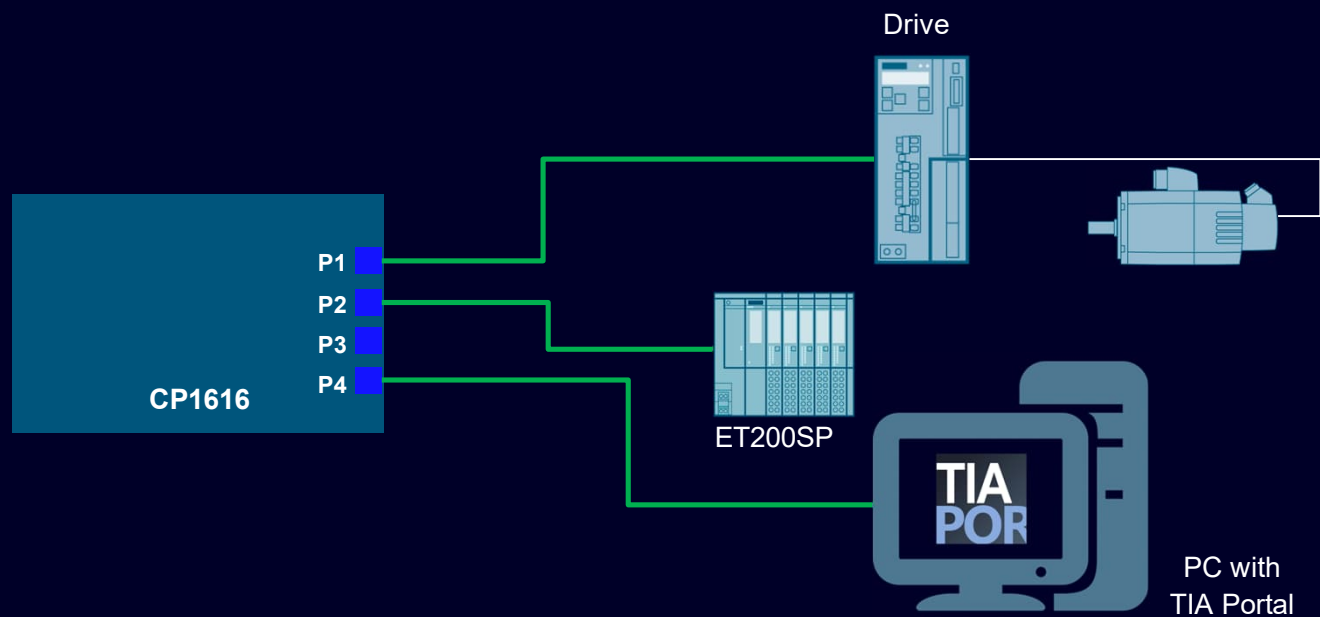
Pass:

- The device works well again after all test cases

Fail:

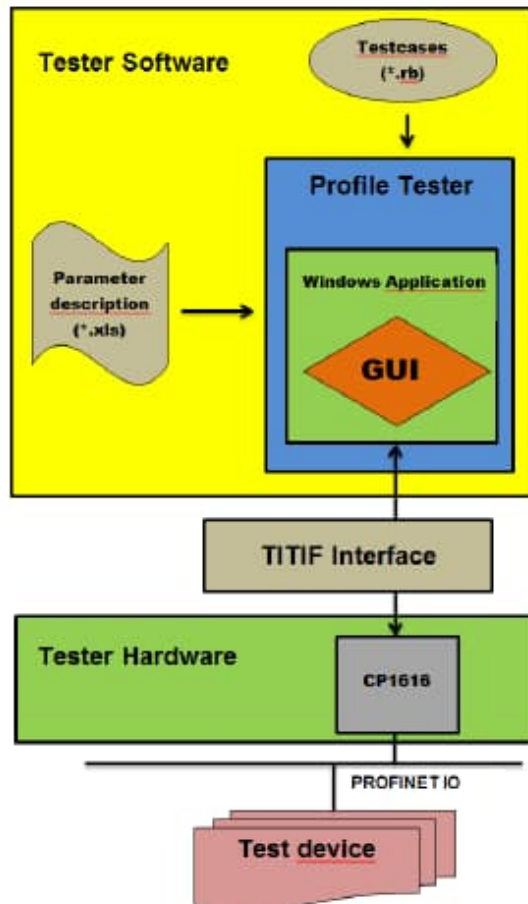
- The DUT resets itself
- A reset is necessary to start the DUT again
- The behavior of the DUT is not according to the test cases (8.3.21 -8.3.83)

Profidrive 测试环境



- ❑ 搭建测试环境，准备ET200SP，至少两个DO分别通过继电器控制驱动器电源和编码器电源
- ❑ 实现规约定义的全部强制内容
- ❑ 至少实现5个可写的非数组型参数
- ❑ 至少实现5个可写的数组型参数
- ❑ 可以通过写参数的方式触发和消除至少两种类型故障（Fault）

Profile Tester



The screenshot shows the **Profile Tester** software interface. The **Menu and Toolbar** at the top includes options like **Test Script**, **Test Queue**, **Telegrams**, **Speed Interpolator**, **Position Interpolator**, and **Device Access**. The **Active DO** dropdown is set to **cu10k13 - L1** with an **OPERATE** button. The **Working area** displays a table of test results.

Name	Path	Result	Failures
Cyclic_Drive_Control_Speed_Control_Mode_01.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_01.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_02.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_02.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_03.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_03.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_04.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_04.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_05.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_05.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_06.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_06.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_07.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_07.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_08.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_08.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_09.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_09.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_10.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_10.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_11.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_11.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_12.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_12.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_13.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_13.rb	Queued	0
Cyclic_Drive_Control_Speed_Control_Mode_14.rb	Script\cyclic Communication\Speed Control mode\Cyclic_Drive_Control_Speed_Control_Mode_14.rb	Queued	0

The **Output window** shows a log of events:

Time	Type	Value
09:56:07:548		Controller mode = 3 (STOP)
09:56:08:000		Controller mode = 4 (OPERATE)
09:56:12:540		Station came cu10k13
09:56:13:427		Sign of Life synchronized, subSet cu10k13.1.1

Annotations on the right side of the screenshot:

- Communication state**: Points to the **Active DO** dropdown.
- Information button**: Points to the **OPERATE** button.
- Watch function overview/ Breakpoints overview/ Error code browser/ Function**: Points to the **Watches** tab in the **View** pane.

Labels for the interface components:

- Script browser/ Parameter browser/ Signal browser**: Points to the **Explore** pane on the left.
- Output window**: Points to the **Output** pane at the bottom.

谢谢！



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聚焦生态 持续共生

西门子 PROFINET 生态论坛

2023年4月12日 · 成都

